

Room Tilt Stick

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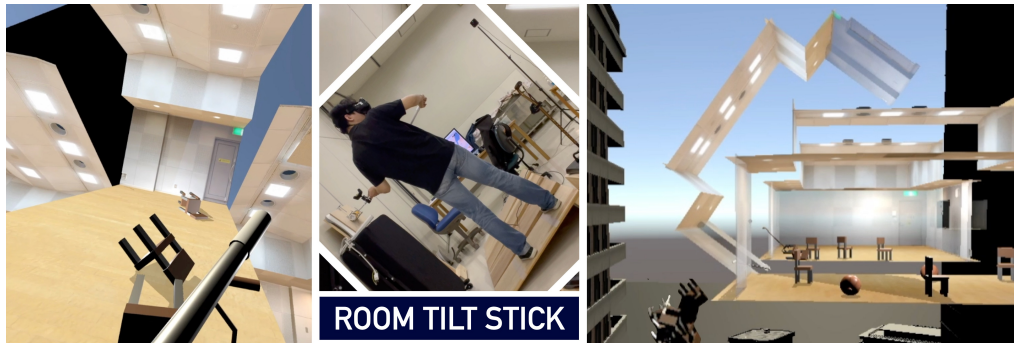


Figure 1: ROOM TILT STICK

ABSTRACT

Room Tilt Stick is an interactive head-mounted display (HMD) system that provides an illusory experience of tilting a room using a stick and of walking on an acutely inclined slope. In the proposed system, the tilting velocity of a three-dimensional modeled room in the HMD's view is correlated with the degree of force transferred to the wall or ground with a specially designed stick. Here a weight scale (i.e., a Wii Balance Board) monitors to what degree the participant's weight is rested against the wall (through a tiltable platform) or ground. After the illusory tilt of the room is completed, participants are instructed to walk on a two meter long wooden slope. The degree of physical inclination is gradually increased according to the walking distance, and the walking slope corresponds to an illusory wall in the HMD's view. The system was tested in our laboratory exhibition, and 87 out of 100 participants reported a strong sensation of the room tilting by pressing the wall or ground with the stick.

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1 INTRODUCTION

A visual scene where a room is spatially tilted and the actors or characters struggle to move around the inclined ground has been

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frequently seen in cartoons and movies. However, creating this type of illusory experience in a head-mounted display (HMD) environment is difficult because such unexpected spatial distortion can easily induce HMD sickness. In this context, it is worth considering how the human cognitive system can endure such a drastic spatial distortion with reference to rare cases where people experience as a subjective visual scene where the room is tilted without tilting themselves. This tilt perception is referred to as "Room Tilt Illusion" (RTI)[Tiliket et al. 1996]. In many cases of RTI, the rotation occurs around the Z axis (forward to backward axis), and the degree of clockwise/counter-clockwise rotation is 90 or 180 degrees[Malis and Guyot 2003][Sierra-Hidalgo et al. 2012]. RTI tends to be triggered by an expected change in motion, such as hitting a bump when driving.

In this paper, referring to the characteristic seen in RTI, we propose Room Tilt Stick, an innovative interactive HMD system that involves an illusory room tilt triggered by shifting of the weight of the physical body.

2 DESIGN

The interactive HMD Room Tilt Stick system consists of four elements: a two meter long wooden inclined slope, two weight scales (i.e., Wii Balance Boards) located at the start and end points of the slope, two types of physical sticks used to exert pressure on the weight scales, and a tiltable platform to support the stick's leaning against the wall.

The interaction starts with the participant's weight resting on a wall by leaning a long rotational stick (i.e., a lint roller) against the wall through the tiltable platform (top of Fig.2). This operation tilts a 3D modeled room in the HMD view in a counter-clockwise direction such that the degree of pressing force transferred to the wall measured by the scale is directly correlated with the tilting

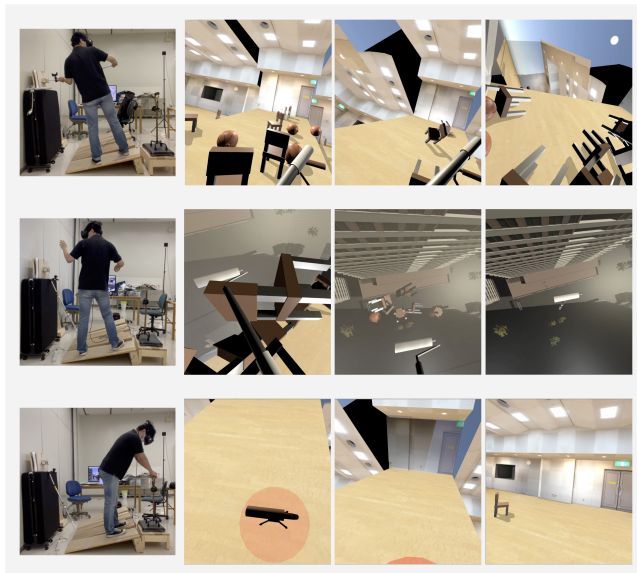


Figure 2: Tilting phase of the system

velocity of the room. In parallel with this rotation, in the HMD view, the room is partially sliced from the original long-rectangular solid, while chairs and balls are thrown outside to drop onto the street far below the inclined floor (middle of Fig.2). This initial phase of the room tilt is completed when the rotation reaches 90 degrees. Then the participants are instructed to walk on the wooden slope while viewing the surroundings through the HMD as shown in Fig.3. Participants perceive that they are walking on the wall and that their bodies are tilted by 90 degrees. They see streets on the left side and the sky among buildings on the right. Note that, initially, the degree of slope inclination is approximately five degrees and gradually increases to a maximum of 18 degrees, according to the walking distance. After experiencing the maximum inclination, the participants are instructed to return to the start point. When reaching this point, they rest their weight on the ground to rotate the sliced room back to the original condition by leaning a four-legged cane against the ground (bottom of Fig.2).

3 REACTIONS IN THE EXHIBITION

This system was opened to the public at a laboratory exhibition in November 2020. Many participants reported experiencing an intense tilting sensation, and while walking in the tilted room they expressed their feelings with impressive fearful voices, as shown in the submitted video. Note that we frequently observed the participants staggering exaggeratedly on the slope. This tends to occur when they turned their head toward the top or bottom of the HMD scene. Considering that the degree of slope inclination was at maximum less than 20 degrees, it is obvious that the subjective sense of self-tilt was strongly distorted through this interaction.

After each two- or three-minute trials with Room Tilt Stick, each participant was asked to rate the illusion strength of tilting the room with the stick (First questionnaire) and the illusion of walking on the wall after the room is tilted (Second questionnaire)

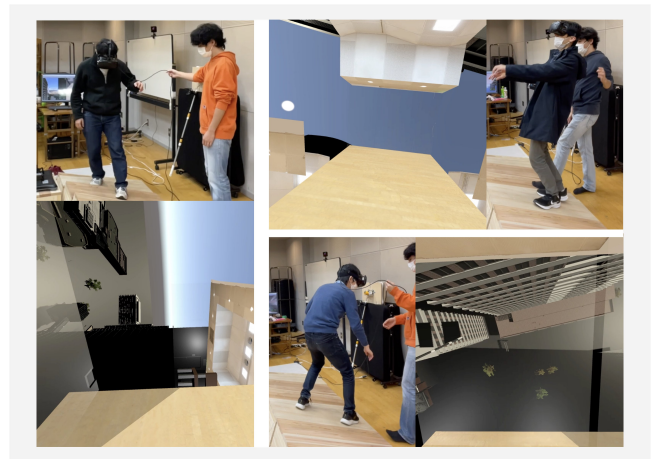


Figure 3: Walking phase of the system

using a seven-point Likert scale (-3 to +3). The result was that 87/56 out of 100 participants strongly agreed with the 1st/2nd statements (rated +2 or more). The mean of the rating score was +2.29/1.48, as shown in Fig.4. These results demonstrate that the proposed interactive system produced a strong and consistent RTI.

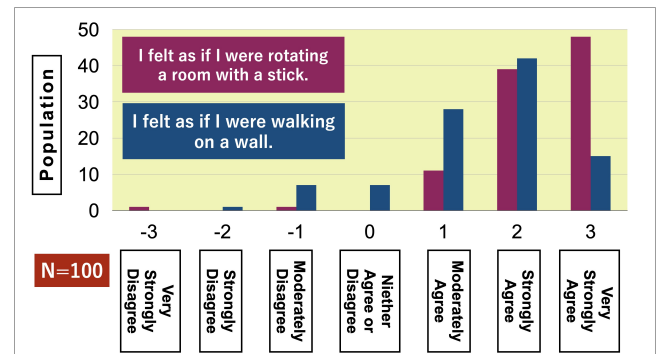


Figure 4: Evaluation for two kinds of subjective reversals

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